

A CONSTRUCTIVE ATTACK ON A RECENT COMBINATORIAL CONJECTURE

Single curves do not force log-concavity

An annular counterexample, a positive transfer matrix,
and Fibonacci-type recurrences

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Abstract

We construct a counterexample to the single-lamination log-concavity conjecture in Section 5 of Kang–Lee–Lim, *Unimodality and Cluster Algebras from Surfaces*, arXiv:2508.04396v3. The surface is an unpunctured annulus with one marked point on each boundary component, and the lamination consists of one unit-weight simple curve. A legal change of triangulation gives the extended exchange matrix with rows $(0, -2)$, $(2, 0)$, and $(-1, 2)$. Three subsequent mutations produce the coefficient polynomial

$$1 + 3q^2 + 2q^3 + 4q^4 + 2q^5 + q^6.$$

The positive consecutive coefficients 3, 2, 4 violate log-concavity and form a strict valley, so deleting internal zeros does not remove the obstruction. Indeed, one fixed triangulation and one fixed curve yield infinitely many nonunimodal examples. We derive a two-parameter family, prove an equivalent linear recurrence, give a nonnegative two-state walk model and a positive finite coefficient formula, identify its row sums with OEIS A001519, and establish exponential and Gaussian asymptotics. The geometric realization is proved from standard surface-cluster results; all finite algebraic certificates are also checked with an included dependency-free exact-arithmetic program. The disproof targets the stated arXiv version; no claim of a completed priority search or independent peer review is made.

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1 The problem and the outcome

For a polynomial $f(q) = \sum_{i=0}^m a_i q^i$ with nonnegative coefficients, *log-concavity* means

$$a_i^2 \geq a_{i-1}a_{i+1} \quad (1 \leq i < m). \quad (1.1)$$

A sequence is *unimodal* if it first weakly increases and then weakly decreases. A strict local valley $a_{i-1} > a_i < a_{i+1}$ therefore disproves unimodality, independently of what happens elsewhere.

The selected problem is Conjecture 1 of Kang–Lee–Lim [1], in the single-lamination setting of their Section 5. Their Definition 20 associates a c -polynomial to a cluster variable by specializing the initial mutable variables to 1 and the frozen variables to q . They conjecture that its coefficient sequence satisfies (1.1) when the lamination has one component. Their Theorem 14 also asserts unimodality on unpunctured surfaces. The construction below contradicts both statements at that level of generality.

Theorem 1.1 (Counterexample and persistence). *There are an unpunctured marked annulus S , a fixed triangulation T , and a fixed lamination L consisting of exactly one unit-weight simple curve such that the c -polynomials of a sequence of simple arcs are $P_n(q)$, where*

$$P_0 = P_1 = 1, \quad P_{n+1} = (1 + q + q^2)P_n - qP_{n-1} \quad (n \geq 1). \quad (1.2)$$

For every $n \geq 3$, the positive consecutive coefficients at degrees 2, 3, 4 violate log-concavity. For every $n \geq 4$, those coefficients form a strict valley. In particular,

$$P_4(q) = 1 + 3q^2 + 2q^3 + 4q^4 + 2q^5 + q^6 \quad (1.3)$$

is neither log-concave nor unimodal, even after all zero coefficients are removed.

The proof has two independent parts. The geometric part verifies that the coefficient row really comes from *one connected curve*, not an arbitrary integral lamination. The algebraic part consists of explicit mutations and coefficient calculations. Both are necessary: a random integer row does not by itself certify connectedness of the corresponding lamination.

1.1 Version and scope of the claim

The source examined is arXiv:2508.04396v3, whose submission history gives 11 February 2026. The arXiv record lists a related publication in the *European Journal of Combinatorics*, volume 136, article 104392 [1]. The final publisher text was not independently inspected. Accordingly, “counterexample to the conjecture” throughout this note means a counterexample to its formulation in that specific arXiv version, with its definitions and exchange rules. The result concerns an annulus, not a polygon-only restriction, and does not contradict the separate results about ordinary rank polynomials of fence or loop-fence posets.

The main mathematical assertion is a proof, rather than an inference from a search bound. The computational appendix corroborates its formulas but is not used to replace an argument for infinitely many indices. The recurrence and transfer-matrix techniques are elementary; we make no separate originality claim for those methods or for the Fibonacci sequence obtained by specialization.

2 The geometric certificate

This section supplies all surface-cluster input needed later. Readers primarily interested in enumerative combinatorics can then work entirely with the recurrences and weighted walks.

2.1 Mutation with one frozen variable

A rank-two geometric seed consists of independent mutable variables x_1, x_2 , a frozen variable q , and an integer matrix

$$\tilde{B} = (b_{ij})_{1 \leq i \leq 3, 1 \leq j \leq 2}$$

whose top 2×2 submatrix is skew-symmetric. Mutation at $k \in \{1, 2\}$ replaces x_k by x'_k according to

$$x_k x'_k = \prod_{i=1}^3 x_i^{[b_{ik}]_+} + \prod_{i=1}^3 x_i^{[-b_{ik}]_+}, \quad x_3 = q, \quad (2.1)$$

where $[z]_+ = \max(z, 0)$. Simultaneously,

$$b'_{ij} = \begin{cases} -b_{ij}, & i = k \text{ or } j = k, \\ b_{ij} + [b_{ik}]_+ [b_{kj}]_+ - [-b_{ik}]_+ [-b_{kj}]_+, & \text{otherwise.} \end{cases} \quad (2.2)$$

These are the usual exchange and matrix-mutation rules; see Fomin–Zelevinsky [4] and Fomin–Thurston [2].

There is just one frozen generator in our construction. Its powers are not new variables. In particular, q^2 remains q^2 after the specialization in Definition 20; it is not renamed q . This distinction matters when a single curve has shear coordinate of absolute value greater than one.

2.2 Two triangles on an annulus

Let S be an annulus with a marked point on each boundary circle and no punctures. A concrete model is a square with its vertical sides identified. The two bottom corners become one marked point and the two top corners become the other. The diagonal and the identified vertical side give two bridging arcs, denoted α_1 and α_2 . Their interiors are disjoint. They cut the annulus into two triangles, neither of which is self-folded.

With a suitable labeling, both triangles contribute the same arrow between these two arcs. Equivalently, in the square model their cyclic side orders give α_2 immediately after α_1 in the signed-adjacency convention in each triangle. Thus the principal exchange matrix is

$$B(T_0) = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}. \quad (2.3)$$

The entry 2 is the sum of two contributions from different triangles. It does not mean that an arc or a lamination has been given multiplicity two.

This annulus lies within the usual marked-surface setup: each boundary component has a marked point, and it has no punctures or self-folded triangles. Arcs and flips have the standard interpretation established by Fomin–Shapiro–Thurston [3] and Fomin–Thurston [2].

2.3 One curve, not merely one coefficient row

Take $L = L_{\alpha_1}$, the elementary lamination following α_1 once, with its endpoints displaced slightly along the boundary in the standard prescribed directions. This is a single simple bridging curve of weight one. Because its endpoints lie on different boundary components, it is not boundary-parallel and does not bound an excluded disk.

The elementary-lamination construction gives the shear row $(1, 0)$ relative to T_0 . This is the one-row part of the principal-coefficient realization in Fomin–Thurston, Definition 17.2 and Proposition 17.3 [2]. We use

only this one curve, not the entire collection of elementary laminations. Hence

$$\tilde{B}_0 = \begin{pmatrix} 0 & 2 \\ -2 & 0 \\ 1 & 0 \end{pmatrix}. \quad (2.4)$$

Keep L fixed and flip α_1 . The shear-coordinate mutation theorem (Fomin–Thurston, Theorem 12.6, or its tagged version Theorem 13.5) says that changing the triangulation changes the extended matrix by (2.2) [2]. Direct calculation gives

$$\tilde{B}_1 = \mu_1(\tilde{B}_0) = \begin{pmatrix} 0 & -2 \\ 2 & 0 \\ -1 & 2 \end{pmatrix}. \quad (2.5)$$

For clarity, the new bottom-right entry is $0 + [1]_+[2]_+ - [-1]_+[-2]_+ = 2$, and the first bottom entry is negated.

Lemma 2.1 (Connected geometric realization). *The matrix (2.5) is the extended exchange matrix of the triangulation $T_1 = \mu_1(T_0)$ and the same one-component, unit-weight lamination $L = L_{\alpha_1}$.*

Proof. The initial realization (2.4) uses the explicit elementary curve just described. The flip changes only the triangulation. The fixed-curve shear-coordinate mutation theorem gives (2.5). Neither the number of connected components of L nor its weight changes under this operation. $\square$

In particular, we have not inferred connectedness from the fact that -1 and 2 are relatively prime. Such an inference would not be a valid general criterion. The actual curve and its legal flip history are the certificate.

2.4 Which variables are set to one?

We now choose T_1 as the *initial triangulation for the c -polynomials*. Write its independent mutable variables as a, b . Every c -polynomial below means specialization of these variables to $a = b = 1$ with q retained.

This is not the same as setting the variables of T_0 to one and then repeatedly resetting newly created variables. To see the distinction explicitly, write the variables of T_0 as u, b . Its first exchange is

$$ua = q + b^2.$$

Thus $a = b = 1$ corresponds to $u = 1 + q$, not to $u = 1$. The birational change of coordinates ensures that a, b remain independent over $\mathbb{Q}(q)$. The general statement being tested allows a chosen initial triangulation, so choosing T_1 is legitimate.

Although L was conveniently described as elementary relative to T_0 , it is not one of the elementary coefficient curves of T_1 : its row there is $(-1, 2)$ rather than a standard basis row. In particular this is not a principal-coefficient rank-polynomial calculation in disguise.

3 The counterexample in three exchanges

Start with (2.5) and the cluster (a, b) . Mutate in directions 2, 1, 2. Denote the successive new cluster variables by v, w, z . The matrices and equations are completely explicit:

$$\tilde{B}_1 = \begin{pmatrix} 0 & -2 \\ 2 & 0 \\ -1 & 2 \end{pmatrix}, \quad bv = a^2 + q^2, \quad (3.1)$$

$$\tilde{B}_2 = \begin{pmatrix} 0 & 2 \\ -2 & 0 \\ 3 & -2 \end{pmatrix}, \quad aw = v^2 + q^3, \quad (3.2)$$

$$\tilde{B}_3 = \begin{pmatrix} 0 & -2 \\ 2 & 0 \\ -3 & 4 \end{pmatrix}, \quad vz = w^2 + q^4. \quad (3.3)$$

After the last exchange the matrix has bottom row $(5, -4)$ and principal part $B(T_0)$. All variables created by these flips correspond to simple arcs on the annulus. No generalized self-intersecting arc or essential closed-loop variable is needed.

3.1 A full Laurent certificate

To avoid relying solely on calculations after specialization, put $A = a^2 + q^2$. The exchanges give

$$v = \frac{A}{b}, \quad w = \frac{A^2 + b^2 q^3}{ab^2}, \quad z = \frac{A^3 + 2Ab^2 q^3 + b^4 q^4}{a^2 b^3}. \quad (3.4)$$

For the last identity, substitute the first two expressions into (3.3). The numerator before cancellation is

$$(A^2 + b^2 q^3)^2 + a^2 b^4 q^4 = A(A^3 + 2Ab^2 q^3 + b^4 q^4),$$

because $q^6 + a^2 q^4 = q^4 A$. This establishes the Laurent expression directly.

Now set $a = b = 1$. The successive polynomials are

$$P_2 = 1 + q^2, \quad (3.5)$$

$$P_3 = 1 + 2q^2 + q^3 + q^4, \quad (3.6)$$

$$P_4 = 1 + 3q^2 + 2q^3 + 4q^4 + 2q^5 + q^6. \quad (3.7)$$

Already P_2 violates (1.1): its coefficients are $(1, 0, 1)$ and $0^2 < 1 \cdot 1$. The stronger example is P_4 . At index 3, its coefficients satisfy

$$([q^3] P_4)^2 = 2^2 = 4 < 3 \cdot 4 = ([q^2] P_4)([q^4] P_4). \quad (3.8)$$

Moreover, $3 > 2 < 4$, so the sequence is not unimodal.

3.2 Why an internal-zero convention cannot rescue the statement

One can additionally require the absence of internal zeros, or instead adopt the more permissive convention of testing only the sequence of nonzero coefficients. Neither choice matters for (3.8). Removing the zero at degree one leaves

$$(1, 3, 2, 4, 2, 1),$$

which still has the strict valley 3, 2, 4 and still violates the corresponding log-concavity inequality. Likewise, multiplying by a monomial merely shifts degrees and does not remove this triple.

One must not generalize the first binomial example carelessly. When $d > 2$, the sequence of $1 + q^d$ can satisfy every *local* inequality (1.1) if internal zeros are allowed, because no zero has two positive neighbors. This is why the positive-consecutive-coefficient obstruction is preferable to an argument based only on a support gap.

4 A two-parameter annular family

The three-exchange calculation is not isolated. The same curve yields arbitrarily large shear coordinates as the triangulation changes.

Lemma 4.1 (Alternating shear-coordinate orbit). *Starting from $\tilde{B}_0$ in (2.4), perform the alternating mutations 1, 2, 1, 2, $\dots$. After $r \geq 0$ mutations, the principal part is $(-1)^r B(T_0)$ and the coefficient row is*

$$\ell_r = \begin{cases} (r+1, -r), & r \text{ even}, \\ (-r, r+1), & r \text{ odd}. \end{cases} \quad (4.1)$$

Every matrix in this orbit is realized by the same one-component lamination L on the same annulus.

Proof. For $r = 0$, the claim is (2.4). Suppose first that r is even. The next mutation is at 1. It negates the first coefficient $r+1$, and changes the second from $-r$ to $-r + 2(r+1) = r+2$. The top matrix changes sign. This gives $(-(r+1), r+2)$, as required at the odd index $r+1$. If r is odd, the next mutation is at 2, and the same calculation with the coordinates interchanged gives $(r+2, -(r+1))$. Induction proves the formulas. The geometric assertion follows by keeping L fixed throughout and applying the shear-coordinate mutation theorem at each flip. $\square$

Fix an integer $d \geq 1$. Choose the triangulation reached after $d-1$ mutations as a new initial seed, and specialize its two mutable variables to 1. Continue forward along the alternating orbit, without further resetting any variable. Write the successive specialized arc variables as $P_2^{(d)}, P_3^{(d)}, \dots$, and set $P_0^{(d)} = P_1^{(d)} = 1$ for the two initial variables in their exchange-chain order.

Proposition 4.2 (Nonlinear recurrence). *The polynomials associated to this construction satisfy*

$$P_{n-1}^{(d)} P_{n+1}^{(d)} = (P_n^{(d)})^2 + q^{d+n-1} \quad (n \geq 1). \quad (4.2)$$

In particular $P_2^{(d)} = 1 + q^d$. All the terms arise from the same fixed curve L , although changing d changes the chosen initial triangulation.

Proof. Immediately before the n th forward exchange, the orbit index is $r = d + n - 2$. The column to be mutated has coefficient $r+1 = d+n-1$ by (4.1). Its nonzero mutable entry is -2 : for even r this is the first column, and for odd r it is the second. Thus the two exchange monomials are q^{d+n-1} and the square of the other mutable variable. The exchanged-out variable is $P_{n-1}^{(d)}$ and the retained variable is $P_n^{(d)}$. Equation (2.1) is exactly (4.2). $\square$

The proof of the next section also shows directly that all these rationally defined expressions are polynomials with nonnegative integer coefficients. Consequently no general positivity theorem is needed for the enumeration that follows.

5 Linearization and positive enumeration

5.1 An invariant that removes the division

Set

$$K_d(q) = 1 + q + q^d. \quad (5.1)$$

Despite the quadratic form of (4.2), the sequence satisfies a second-order linear recurrence with a fixed polynomial coefficient.

Theorem 5.1 (Linear recurrence and generating functions). *For every $d \geq 1$,*

$$P_0^{(d)} = P_1^{(d)} = 1, \quad P_{n+1}^{(d)} = K_d(q)P_n^{(d)} - qP_{n-1}^{(d)} \quad (n \geq 1). \quad (5.2)$$

The ordinary generating functions are

$$\sum_{n \geq 0} P_n^{(d)}(q)t^n = \frac{1 - (q + q^d)t}{1 - (1 + q + q^d)t + qt^2}, \quad (5.3)$$

$$\sum_{N \geq 0} P_{N+1}^{(d)}(q)t^N = \frac{1 - qt}{1 - (1 + q + q^d)t + qt^2}. \quad (5.4)$$

Proof. Temporarily work in the rational function field and abbreviate $P_n = P_n^{(d)}$. Define

$$K_n = \frac{P_{n+1} + qP_{n-1}}{P_n} \quad (n \geq 1).$$

For $n \geq 2$, multiplication by $P_n P_{n-1}$ gives

$$\begin{aligned} P_n P_{n-1} (K_n - K_{n-1}) &= P_{n+1} P_{n-1} - P_n^2 - q(P_n P_{n-2} - P_{n-1}^2) \\ &= q^{d+n-1} - q q^{d+n-2} = 0. \end{aligned}$$

Thus $K_n = K_1 = P_2 + q = 1 + q + q^d$. The rational functions in question are nonzero: for positive q the original exchange recursion gives positive values. Rearranging $K_n = K_d$ proves (5.2).

Conversely, define P_n by (5.2) and put $D_n = P_{n-1}P_{n+1} - P_n^2$. Direct substitution yields

$$D_n = qD_{n-1} \quad (n \geq 2), \quad D_1 = P_2 - 1 = q^d.$$

Hence $D_n = q^{d+n-1}$, recovering the nonlinear identity. In particular the linear recurrence is not merely a heuristic fit to initial terms.

Finally, multiplying the generating series by $1 - K_d t + qt^2$ leaves only its constant and linear coefficients. These are 1 and $1 - K_d = -(q + q^d)$, respectively, giving (5.3). Subtracting its constant term and dividing by t gives (5.4). $\square$

5.2 A nonnegative transfer matrix

The negative term in (5.2) conceals positivity, but a 2×2 transfer matrix makes it transparent. Define

$$M_d(q) = \begin{pmatrix} 1 + q^d & q^d \\ q & q \end{pmatrix}. \quad (5.5)$$

Proposition 5.2 (Positive walk model). *For every $n \geq 1$,*

$$P_n^{(d)}(q) = (M_d(q)^{n-1})_{11}. \quad (5.6)$$

Equivalently, $P_n^{(d)}$ counts length- $n - 1$ walks that start and end at state A in the following weighted directed multigraph:

Transition	Number of edges	Integer edge weights
$A \rightarrow A$	2	0 and d
$A \rightarrow B$	1	d
$B \rightarrow A$	1	1
$B \rightarrow B$	1	1

A walk of total edge weight k contributes q^k . In particular, $P_n^{(d)} \in \mathbb{Z}_{\geq 0}[q]$.

Proof. The trace of M_d is K_d and its determinant is q . Its characteristic identity is therefore $M_d^2 - K_d M_d + qI = 0$. The $(1, 1)$ entries of its powers satisfy (5.2), with initial entries 1 and $1 + q^d$. These agree with P_1 and P_2 , proving (5.6). The entries of M_d are the generating polynomials for the displayed transitions, so ordinary matrix multiplication proves the walk interpretation. Each counted walk has a nonnegative integer multiplicity. $\square$

For $d \geq 2$ and $n \geq 2$, this also gives

$$\deg P_n^{(d)} = d(n - 1), \quad [q^0]P_n^{(d)} = [q^{d(n-1)}]P_n^{(d)} = 1. \quad (5.7)$$

Indeed, only the all-zero A -loop walk has weight zero. A walk of length $n - 1$ has weight at most $d(n - 1)$; attaining this upper bound requires every edge to have weight d . A walk returning from B would use an edge of weight $1 < d$, so the only maximizing walk consists of the positive A -loop at every step. For $d = 1$ the upper-degree uniqueness argument does not apply, which is why that case is excluded from the monicity assertion.

5.3 Excursions and a positive finite formula

A closed walk at A decomposes uniquely into individual A -loops and excursions from A through B back to A . If an excursion has $b \geq 0$ intermediate B -loops, its length is $b + 2$ and its weight is $d + b + 1$. Its generating series is thus

$$\sum_{b \geq 0} t^{b+2} q^{d+b+1} = \frac{q^{d+1} t^2}{1 - qt}.$$

A closed walk is a sequence of these blocks and the two possible A -loops. This gives a second derivation of (5.4):

$$\sum_{N \geq 0} P_{N+1}^{(d)} t^N = \frac{1}{1 - (1 + q^d)t - q^{d+1}t^2/(1 - qt)}. \quad (5.8)$$

Theorem 5.3 (A subtraction-free finite expansion). *For $N \geq 0$,*

$$\begin{aligned} P_{N+1}^{(d)}(q) &= (1 + q^d)^N \\ &+ \sum_{j=1}^{\lfloor N/2 \rfloor} \sum_{b=0}^{N-2j} \binom{N-j-b}{j} \binom{b+j-1}{j-1} q^{j(d+1)+b} (1 + q^d)^{N-2j-b}. \end{aligned} \quad (5.9)$$

Proof. The first term counts walks without excursions. Otherwise let j be the number of excursions, let b be the total number of internal B -loops, and let $a = N - 2j - b$ be the number of A -loops. There are $\binom{a+j}{j}$ ways to position the j ordered excursions among the a individual A -loops. There are $\binom{b+j-1}{j-1}$ ways to distribute b internal loops among the j excursions. Their combined weight is $j(d+1) + b$, and the A -loops contribute $(1+q^d)^a$. Summing gives (5.9), since $a+j = N-j-b$. $\square$

Thus these are ordinary weighted counting polynomials for a fixed finite state system, in addition to being specializations of surface cluster variables. This makes the recurrence and coefficient triangle accessible without any cluster-algebra terminology.

6 Infinitely many positive-coefficient obstructions

Let $a_{n,k}^{(d)} = [q^k]P_n^{(d)}$. The beginning of each row can be read directly from the walk model.

Proposition 6.1 (Low-degree coefficients). *For $d \geq 2$ and $n \geq 1$,*

$$a_{n,0}^{(d)} = 1, \quad a_{n,k}^{(d)} = 0 \quad (1 \leq k < d), \quad (6.1)$$

$$a_{n,d}^{(d)} = n - 1, \quad (6.2)$$

$$a_{n,d+s}^{(d)} = \max(n - 1 - s, 0) \quad (1 \leq s \leq d - 1), \quad (6.3)$$

$$a_{n,2d}^{(d)} = \binom{n-1}{2} + \max(n - d - 1, 0). \quad (6.4)$$

Here $\binom{0}{2} = \binom{1}{2} = 0$.

Proof. Put $N = n - 1$. The smallest positive A -loop weight is d and the smallest excursion weight is $d + 1$. This proves (6.1). A weight- d walk must contain one positive A -loop and otherwise zero loops, giving N choices and (6.2).

For weight $d + s < 2d$, there cannot be two positive A -loops, two excursions, or one positive A -loop together with an excursion. The only possibility is one excursion with $s - 1$ internal B -loops. It occupies $s + 1$ steps and can be placed in $N - s$ positions if $N \geq s + 1$, and in no positions otherwise. This gives (6.3).

At weight $2d$, there are exactly two possibilities. Two positive A -loops can be chosen in $\binom{N}{2}$ ways. Alternatively, a single excursion with $d - 1$ internal B -loops occupies $d + 1$ steps and has $\max(N - d, 0)$ possible positions. Their sum is (6.4). $\square$

Corollary 6.2 (Every $d \geq 2$ supplies a positive triple). *At $n = d + 1$, the coefficients of $P_n^{(d)}$ at consecutive degrees $2d - 2, 2d - 1, 2d$ are*

$$\left(2, 1, \binom{d}{2}\right). \quad (6.5)$$

They violate log-concavity for every $d \geq 2$. For $d \geq 3$ they also form a strict local valley.

Proof. For $d = 2$, the first entry follows from (6.2); for $d \geq 3$ it follows from (6.3) with $s = d - 2$. The remaining entries follow from (6.3) and (6.4). The log-concavity inequality would require $1 \geq 2\binom{d}{2} = d(d-1)$, which is false for every $d \geq 2$. For $d \geq 3$, both neighbors of the middle entry exceed one. $\square$

More is true without varying the initial seed at all.

Proof of theorem 1.1. Use the fixed initial seed T_1 , so $d = 2$. The geometric realization and recurrence follow from theorems 2.1, 4.2 and 5.1. For $n \geq 3$, theorem 6.1 gives the positive triple

$$\left(a_{n,2}^{(2)}, a_{n,3}^{(2)}, a_{n,4}^{(2)}\right) = \left(n-1, n-2, \binom{n-1}{2} + n-3\right). \quad (6.6)$$

Write $m = n - 3 \geq 0$. The neighbor product minus the middle square equals

$$\begin{aligned} (n-1) \left(\binom{n-1}{2} + n-3 \right) - (n-2)^2 \\ = \frac{m^3 + 5m^2 + 8m + 2}{2} > 0. \end{aligned} \quad (6.7)$$

Therefore log-concavity fails for every $n \geq 3$. If $n \geq 4$, the first entry in (6.6) exceeds the second by one, while the third exceeds the second by $\binom{n-1}{2} - 1 > 0$. Thus the triple is a strict valley. The expression for P_4 was calculated in section 3. The degrees $2(n-1)$ strictly increase with n , so these polynomials are indeed infinitely many distinct examples. $\square$

For reference, the first coefficient rows for $d = 2$ are shown below. Blank entries are outside the degree range, not additional coefficients.

$n \backslash k$	0	1	2	3	4	5	6	7	8	9	10	$P_n(1)$
0	1											1
1	1											1
2	1	0	1									2
3	1	0	2	1	1							5
4	1	0	3	2	4	2	1					13
5	1	0	4	3	8	7	7	3	1			34
6	1	0	5	4	13	14	20	16	11	4	1	89

7 Closed forms, Fibonacci numbers, and asymptotics

7.1 Exact algebraic form and row sums

Fix $d \geq 1$ and $q > 0$, and define

$$\Delta_d(q) = \sqrt{K_d(q)^2 - 4q}, \quad \lambda_{\pm}(q) = \frac{K_d(q) \pm \Delta_d(q)}{2}. \quad (7.1)$$

Since $K_d > 1 + q \geq 2\sqrt{q}$, the roots are real, distinct, and positive. The characteristic polynomial $x^2 - K_d x + q$ is negative at both $x = 1$ and $x = q$. Therefore

$$0 < \lambda_- < \min(1, q) \leq \max(1, q) < \lambda_+. \quad (7.2)$$

Proposition 7.1 (Closed form). *For every $n \geq 0$ and $q > 0$,*

$$P_n^{(d)}(q) = \frac{(1 - \lambda_-)\lambda_+^n + (\lambda_+ - 1)\lambda_-^n}{\lambda_+ - \lambda_-}. \quad (7.3)$$

At $q = 1$, independently of d ,

$$P_0^{(d)}(1) = 1, \quad P_n^{(d)}(1) = F_{2n-1} \quad (n \geq 1), \quad (7.4)$$

where $F_0 = 0$, $F_1 = 1$, and $F_{r+1} = F_r + F_{r-1}$.

Proof. The right side of (7.3) satisfies the linear recurrence, because both characteristic roots do. Its values at $n = 0$ and $n = 1$ are one, so uniqueness proves the formula. At $q = 1$, the recurrence becomes $P_{n+1} = 3P_n - P_{n-1}$, with $P_0 = P_1 = 1$. The odd-indexed Fibonacci numbers satisfy this recurrence and the required initial values for $n \geq 1$. Alternatively, substitute $\lambda_+ = \varphi^2$ and $\lambda_- = \varphi^{-2}$, where $\varphi = (1 + \sqrt{5})/2$, into (7.3) and use Binet's formula. $\square$

The row sums are exactly OEIS A001519 with its existing indexing:

$$1, 1, 2, 5, 13, 34, 89, 233, 610, 1597, \dots$$

That entry defines the sequence by the same recurrence and initial values, and identifies it as an odd-indexed Fibonacci subsequence [5]. Thus no new OEIS identifier is being claimed for these totals. The family provides a weight refinement of that established sequence.

There is a useful interpretation of the independence from d . At $q = 1$,

$$M_d(1) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

is independent of d . The set of walks is the same for every parameter; only the statistic assigned to each walk changes. Coefficient-shape phenomena therefore change while the row sums do not.

7.2 Exponential asymptotics at fixed positive q

Define

$$A_d(q) = \frac{1 - \lambda_-(q)}{\Delta_d(q)}, \quad C_d(q) = \frac{\lambda_+(q) - 1}{\Delta_d(q)}.$$

Both are positive by (7.2). Equation (7.3) gives

$$P_n^{(d)}(q) = A_d(q) \lambda_+(q)^n \left(1 + \frac{C_d(q)}{A_d(q)} \left(\frac{\lambda_-(q)}{\lambda_+(q)} \right)^n \right). \quad (7.5)$$

This is an exact two-term expansion, and hence in particular an exponentially accurate leading asymptotic. For fixed d , it is uniform when q ranges over any compact subset of $(0, \infty)$: the positive continuous quantities stay bounded away from their forbidden limits, and the root ratio has a maximum strictly less than one on that compact set.

At $q = 1$, the exact expression is especially simple:

$$P_n^{(d)}(1) = \frac{\varphi^{2n-1} + \varphi^{-(2n-1)}}{\sqrt{5}} = \frac{\varphi^{2n-1}}{\sqrt{5}} \left(1 + \varphi^{-4n+2} \right) \quad (n \geq 1). \quad (7.6)$$

7.3 A Gaussian limit despite persistent non-log-concavity

Choose uniformly among all length- $n - 1$ walks from A to A , counting the two A -loops as distinct edges. Let $W_n^{(d)}$ be the total weight. Its probability generating function is

$$\mathbb{E}[q^{W_n^{(d)}}] = \frac{P_n^{(d)}(q)}{P_n^{(d)}(1)}. \quad (7.7)$$

Theorem 7.2 (Mean, variance, and central limit). *For each fixed $d \geq 1$, as $n \rightarrow \infty$,*

$$\mathbb{E}W_n^{(d)} = \mu_d n + O(1), \quad \mu_d = \frac{1}{2} + \frac{2d-1}{2\sqrt{5}}, \quad (7.8)$$

$$\text{Var}(W_n^{(d)}) = \sigma_d^2 n + O(1), \quad \sigma_d^2 = \frac{\sqrt{5}(2d^2 - 2d + 3)}{25}. \quad (7.9)$$

Furthermore,

$$\frac{W_n^{(d)} - \mu_d n}{\sqrt{\sigma_d^2 n}} \Rightarrow N(0, 1). \quad (7.10)$$

Proof. The discriminant in (7.1) is nonzero at $q = 1$. Choose analytic branches of the roots in a complex neighborhood of that point. Their moduli remain separated in a sufficiently small neighborhood, and A_d is nonzero there. If s is sufficiently close to zero, put $H(s) = \log \lambda_+(e^s)$, choosing the analytic logarithm through its positive real value at zero. From (7.5), uniformly in a smaller neighborhood,

$$\log \frac{P_n^{(d)}(e^s)}{P_n^{(d)}(1)} = n(H(s) - H(0)) + \log \frac{A_d(e^s)}{A_d(1)} + O(\rho^n), \quad 0 < \rho < 1. \quad (7.11)$$

Analyticity and a smaller disk also permit differentiation of the exponentially small remainder. Thus the first two cumulants have leading coefficients $H'(0)$ and $H''(0)$.

We compute them explicitly rather than leaving them as derivatives of a root. The characteristic equation becomes

$$e^{H(s)} + e^{s-H(s)} = 1 + e^s + e^{ds}.$$

Put $L = (3 + \sqrt{5})/2$, $l = (3 - \sqrt{5})/2$, $\mu = H'(0)$, and $\nu = H''(0)$. The first and second derivatives at zero give

$$\begin{aligned} L\mu + l(1 - \mu) &= 1 + d, \\ L(\mu^2 + \nu) + l((1 - \mu)^2 - \nu) &= 1 + d^2. \end{aligned}$$

Since $L - l = \sqrt{5}$, the first equation yields (7.8). Substituting that value into the second yields

$$\nu = \frac{1 + d^2 - L\mu^2 - l(1 - \mu)^2}{\sqrt{5}} = \frac{\sqrt{5}(2d^2 - 2d + 3)}{25} > 0.$$

This proves the mean and variance formulas.

For the limit law, expand $H(s) - H(0) = \mu_d s + \frac{1}{2}\sigma_d^2 s^2 + O(s^3)$ in (7.11), set $s = iu/\sqrt{\sigma_d^2 n}$ for fixed real u , and subtract the centering term $iu\mu_d n/\sqrt{\sigma_d^2 n}$. The logarithm of the resulting characteristic function tends to $-u^2/2$; the analytic prefactor contributes $O(n^{-1/2})$ and the cubic remainder does likewise. The continuity theorem for characteristic functions proves (7.10). $\square$

For the fixed-seed counterexample family $d = 2$, the constants are

$$\mu_2 = \frac{1}{2} + \frac{3\sqrt{5}}{10}, \quad \sigma_2^2 = \frac{7\sqrt{5}}{25}.$$

There is no contradiction between this Gaussian limit and the coefficient valleys proved earlier. The central limit theorem describes weight on the scale n with fluctuations of order $\sqrt{n}$. Our persistent obstruction occurs at the fixed degrees 2, 3, 4, far into the lower tail. Those coefficients grow only polynomially in n , whereas the row sum grows exponentially. A Gaussian weak limit neither asserts local log-concavity nor controls every fixed tail coefficient.

8 Interpretation, verification, and remaining questions

8.1 What the counterexample does and does not show

The decisive distinction is between *one connected curve* and *at most one contributing crossing at each mutable arc*. The former does not imply the latter on an annulus. The coefficient row $(-1, 2)$ is obtained from one curve by one flip, and (4.1) exhibits unbounded shear coordinates with no change in the number of curve components. The resulting exchange relation contains q^2 , then q^3 , then q^4 .

This identifies a concrete obstruction to extending arguments based only on unit coefficient shifts to arbitrary unpunctured surfaces. It is unnecessary to diagnose every line of an existing proof: the explicit example already settles the general formulation. The mechanism, however, explains why tests confined to polygons could miss it.

The natural residual question is the polygon-restricted formulation: does log-concavity hold for every single lamination in an unpunctured polygon, with the same geometric-coefficient specialization? This note does not answer that question. Nor does it characterize exactly which triples (S, T, L) produce log-concave polynomials. A further possible restriction is to bound the shear-coordinate multiplicities relevant to an expansion, but the sufficiency of such a condition would require a new proof; it is not asserted here.

The annular construction is deliberately narrow enough to be audited. It uses a rank-two exchange matrix, a single unit-weight bridge, legal flips, and ordinary integer arithmetic. It does not rely on a large experimental search, a floating-point inequality, or a numerical guess about a lamination's connectedness.

8.2 Exact computational verification

The supplied `verify.py` uses the Python standard library and represents polynomials as tuples of integers in increasing degree. Polynomial division checks both integrality of the quotient and vanishing of the remainder. It implements the full extended-matrix mutation rule separately from the linear and nonlinear recurrences and separately from the walk update. The following checks were actually executed:

Check	Number of cases
Shear-orbit matrices, with both mutation involutions	201
Linear/walk agreement, positivity, and Fibonacci row sums	972
Nonlinear recurrence and actual seed mutations	396
Positive finite expansion against the recurrence	156
Fixed-seed log-concavity failures, $3 \leq n \leq 160$	158
Fixed-seed strict valleys, $4 \leq n \leq 160$	157

The 972 cases are $1 \leq d \leq 12$ and $0 \leq n \leq 80$. The 396 nonlinear/mutation cases use the same d range and $0 \leq n \leq 32$. The finite-expansion comparison uses $0 \leq n \leq 12$. All comparisons are exact. The detailed low-degree formulas, degree, monicity, and the general triple (6.5) are also checked in their stated ranges.

These computations do not machine-formalize the geometric theorem or certify all n . The article supplies the geometric reasoning and the induction and counting proofs that make the conclusions unconditional for the stated infinite families. No Lean formalization or external computer-algebra certification is claimed.

8.3 Reproducibility and research status

To rerun the checks and regenerate the data, use Python 3.10 or later:

```
python verify.py --write-data
```

The program writes a matrix-and-polynomial certificate, a coefficient table for $d = 2$ through $n = 40$, Fibonacci row sums through $n = 100$, and a JSON verification report. The archive also includes the LaTeX source, this PDF, a source-audit note, and build instructions. The final proof does not depend on reproducing a random search.

As of the research date, the arXiv record inspected still presented the target as a conjecture. Targeted searches by title and identifier did not locate a correction addressing this annular example. That is not an exhaustive priority determination. Before preparing a journal submission, the appropriate next verification is to compare the final publisher version and seek expert review of the interpretation of the geometric coefficients. The counterexample and all algebraic claims made here are stated explicitly enough for that review to be short and decisive.

A A minimal independent check of the fixed-seed family

The following small program uses only coefficientwise recurrence updates and is included as `minimal_check.py`. It is not the main verifier, since it does not implement the surface matrix. Its purpose is to make the central coefficient claim reproducible with very little code.

```
def add(a, b):
    c = [0] * max(len(a), len(b))
    for i, v in enumerate(a):
        c[i] += v
    for i, v in enumerate(b):
        c[i] += v
    while len(c) > 1 and c[-1] == 0:
        c.pop()
    return c

def shift(a, k):
    return [0] * k + a

P = [[1], [1]]
for n in range(1, 160):
    next_p = add(add(P[n], shift(P[n], 1)), shift(P[n], 2))
    next_p = add(next_p, [-v for v in shift(P[n-1], 1)])
    P.append(next_p)

assert P[4] == [1, 0, 3, 2, 4, 2, 1]
assert all(p[3]**2 < p[2]*p[4] for p in P[3:])
assert all(p[2] > p[3] < p[4] for p in P[4:])
print("PASS: exact fixed-seed checks through n=160")
```

B Certificate at a glance

The essential proof data can be summarized without any asymptotic or generating-function machinery:

Surface	Annulus, one marked point per boundary, no punctures.
Reference triangulation	Two bridges forming two triangles; principal matrix $\begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$.
Lamination	One elementary curve along the first bridge, with weight one; reference shear row $(1, 0)$.
Geometric certification	Flip reference arc 1, keeping the same curve; new coefficient row $(-1, 2)$.
Chosen initial variables	The two mutable variables a, b of this new seed; set $a = b = 1$ only when forming c -polynomials.
Subsequent mutations	2, 1, 2.
Exchange equations	$bv = a^2 + q^2$, $aw = v^2 + q^3$, $vz = w^2 + q^4$.
Resulting polynomial	$1 + 3q^2 + 2q^3 + 4q^4 + 2q^5 + q^6$.
Violation	$2^2 < 3 \cdot 4$, using positive coefficients at degrees 2, 3, 4.

References

- [1] W. Kang, K. J. Lee, and E. Lim, *Unimodality and Cluster Algebras from Surfaces*, arXiv:2508.04396v3, 11 February 2026. Target statements: Definition 20, Theorem 14, and Conjecture 1 in Section 5. <https://arxiv.org/abs/2508.04396v3>; <https://arxiv.org/html/2508.04396v3>. The arXiv record lists the related journal publication *European Journal of Combinatorics* **136** (2026), 104392, <https://doi.org/10.1016/j.ejc.2026.104392>. Version-specific text and metadata consulted 19 September 2026.
- [2] S. Fomin and D. Thurston, *Cluster algebras and triangulated surfaces. Part II: Lambda lengths*, *Memoirs of the American Mathematical Society* **255** (2018), no. 1223, v+97 pp. <https://arxiv.org/abs/1210.5569>. Especially Definitions 12.5 and 17.2, Theorems 12.6 and 13.5, and Proposition 17.3.
- [3] S. Fomin, M. Shapiro, and D. Thurston, *Cluster algebras and triangulated surfaces. I. Cluster complexes*, *Acta Mathematica* **201** (2008), 83–146. <https://arxiv.org/abs/math/0608367>.
- [4] S. Fomin and A. Zelevinsky, *Cluster algebras. I. Foundations*, *Journal of the American Mathematical Society* **15** (2002), 497–529. <https://arxiv.org/abs/math/0104151>.
- [5] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A001519, $a(0) = a(1) = 1$, $a(n) = 3a(n-1) - a(n-2)$. <https://oeis.org/A001519>. Consulted 19 September 2026.